



Tecnológico de Monterrey

Software Requirements Specification

Software Construction and Decision Making

Grupo 501

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Software Requirements Specification

a. Functional requirements

1. The map should change every time I start a new game
2. I want cards to feel integrated into the game style for the combats
3. There must be multiple levels through the game
4. There should be different character options
5. There must be a timer bar system for each combat
6. I would like to save the progress
7. There should be math mechanics to fight each enemy
8. Math problems must be solved for each combat
9. There must be random chests along the game
10. There should be attack cards to damage enemies
11. There should be defense cards to reduce incoming damage
12. There must be special skill cards with unique abilities
13. Health bars should display the player and enemy HP
14. There should be a functional main menu
15. There must be a reward system after each combat
16. Enemies should have unique abilities and behaviors
17. There must be boss battles at the end of each world
18. The type of math operations should vary depending on the map level
19. Players should be able to choose their deck cards before each combat
20. There should be visual effects for attacks and defense actions
21. There must be feedback for correct answers
22. There must be feedback for incorrect answers
23. There should be audio settings for the player
24. There must be a credits screen
25. There should be a progression system throughout the game
26. Players should retain part of their progress after losing a run
27. Players should be able to choose the game difficulty.
28. Players should be able to select preferred math operations in order to practice a specific type.
29. There must be a pause menu during gameplay, excepting if the player is in combat with an enemy

b. Non-functional requirements

1. There should be sound effects for gameplay actions
2. There should be background music during gameplay
3. There must be an instructions screen for the player

c. User Stories:

User story #1 Random Map Generation

As a user

i want the map to change every time I start a new game

so that I can have a different experience in every run

Validation:

- Verify that map routes change between runs.
- Confirm enemies and chests appear in different positions.
- Check that players can choose different paths.

Priority: high

Estimation: 8 hours

User story #2 Pixel styled Cards

As a user

i want cards to feel integrated into the game environment and combat

so that the gameplay feels more immersive.

Validation:

- Verify cards appear naturally during combat.
- Confirm cards have a good style.
- Check that players can interact with cards easily.

Priority: High

Estimation: 6 hrs

User story #3 Multiple levels

As a user

i want multiple levels and worlds throughout the game

so that I can experience progression and new challenges.

Validation: • Verify the existence of at least 3 different maps. • Confirm enemies and difficulty change per level. • Check transitions between levels.	Priority: High Estimation: 10 hrs
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User story #4 Character Selection As a user i want to choose between different character types or skins so that I can personalize my gameplay experience.	
Validation: • Verify multiple selectable characters exist. • Confirm selected skins appear during gameplay. • Check functionality of the character selection screen.	Priority: Medium Estimation: 4 hrs

User story #5 Timer Bar System As a user i want a timer system during math challenges so that the game feels more challenging and engaging.	
Validation: • Verify timer functionality. • Confirm timer color changes correctly. • Check reduced effectiveness when answers are slower.	Priority: High Estimation: 6 hrs

User story #6 Save System

As a user

i want to save my game progress

so that I can continue playing in future sessions, and keep track of my improvement.

Validation:

- Verify game progress is stored correctly.
- Confirm saved data loads properly.
- Check persistence of cards and level progression.

Priority: High**Estimation:** 10 hrs**User story #7 Combat System**

As a user

i want to fight enemies using cards and math mechanics

so that I can progress through the game.

Validation:

- Verify card activation works properly.
- Confirm enemies receive damage correctly.
- Check combat turn system functionality.

Priority: High**Estimation:** 14 hrs**User story #8 Math problem system**

As a user

i want to solve math problems during combat

so that I can activate attacks and abilities.

Validation:

- Verify math problems are generated correctly.
- Confirm answer validation works.
- Check progressive difficulty scaling.

Priority: High

Estimation: 10 hrs

User story #9 Random chests

As a user

i want to encounter random chests

so that each run feels unique and unpredictable.

Validation:

- Verify chest spawning.
- Confirm random events trigger correctly.
- Check reward and trap functionality.

Priority: High

Estimation: 10 hrs

User story #10 Attack Cards

As a user

i want to use attack cards against enemies

so that I can defeat enemies faster.

Validation:

- Verify attack cards deal damage.
- Confirm attack animations trigger correctly.
- Check enemy HP reduction.

Priority: High

Estimation: 10 hrs

User story #11 Defense Cards

As a user

i want to use defensive cards

so that I can reduce incoming damage and protect myself.

Validation:

- Verify defense effects work properly.
- Confirm damage reduction applies correctly.
- Check effect duration.

Priority: High

Estimation: 10 hrs

User story #12 Skill Cards

As a user

i want to unlock special skill cards

so that I can create stronger strategies.

Validation:

- Verify skill cards unlock properly.
- Confirm special effects function correctly.
- Check deck restrictions for skill cards.

Priority: High

Estimation: 10 hrs

User story #13 Health Bars

As a user

i want to see health bars during combat

so that I can track battle progress.

Validation:

- Verify HP bars update correctly.
- Confirm health bars are visible.

Priority: High

Estimation: 6 hrs

• Check combat synchronization.	
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User story #14 Background Music As a user i want background music during gameplay so that the game feels more immersive and enjoyable.	
Validation: • Verify music playback works. • Confirm different music themes per scenario. • Check adjustable volume settings.	Priority: Medium Estimation: 4 hrs

User story #15 Sound Effects As a user i want sound effects during gameplay so that I receive immediate feedback from my actions.	
Validation: • Verify combat sound effects. • Confirm correct and incorrect answer sounds. • Check audio settings functionality.	Priority: Low Estimation: 6 hrs

User story #16 Instructions Screen As a user i want access to an instructions screen so that I can understand how to play the game.	
Validation:	Priority: High Estimation: 10 hrs

• Verify access from the main menu. • Confirm explanation of controls. • Check explanations for cards and timer system.	
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User story #17 Main menu As a user i want a functional main menu so that I can navigate the game easily.	
Validation: • Verify all buttons function properly. • Confirm access to options and credits. • Check screen transitions.	Priority: High Estimation: 8 hrs

User story #18 Reward System As a user i want new cards after winning battles so that I feel progression and achievement.	
Validation: • Verify rewards are granted correctly. • Confirm new cards unlock. • Check level progression functionality.	Priority: High Estimation: 8 hrs

User story #19 Enemies with Unique Skills As a user	
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i want enemies to have unique abilities
so that combat feels more varied and interesting.

Validation:

- Verify unique enemy abilities.
- Confirm special effects activate correctly.
- Check gameplay balance.

Priority: High

Estimation: 14 hrs

User story #20 Boss Battles

As a user

i want boss fights at the end of each world
so that I can experience more challenges.

Validation:

- Verify boss encounters trigger correctly.
- Confirm unique boss mechanics.
- Check rewards after defeating bosses.

Priority: High

Estimation: 14 hrs

User story #21 Different Map Difficulty

As a user

i want the game difficulty to increase gradually
so that I continue feeling challenged as I progress.

Validation:

Validation:

- Verify card activation difficulty scaling.
- Confirm more complex math problems appear.

Priority: High

Estimation: 8 hrs

User story #22 Combat Deck of cards

As a user

i want to choose my deck for each combat with an enemy
so that i can change my strategy and defeat the enemy.

Validation:**Validation:**

- Verify available cards for the combat.
- Confirm the selection of cards.

Priority: High**Estimation:** 8 hrs**User story #23 Visual Feedback**

As a user

i want visual effects during attacks and defense actions
so that combat actions feel satisfying and clear.

Validation:

- Verify combat animations trigger properly.
- Confirm different effects for card types.
- Check animation smoothness.

Priority: High**Estimation:** 8 hrs**User story #24 Correct Answer Feedback**

As a user

i want feedback when I answer correctly
so that I know my action in the combat succeeded.

Validation:

- Verify positive feedback messages.

Priority: low**Estimation:** 1 hrs

• Confirm success sounds trigger. • Check corresponding effects apply correctly.	
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User story #25 Incorrect Answer Feedback

As a user

i want feedback when I answer incorrectly

so that I understand if I made a mistake.

Validation:

- Verify incorrect answer notifications.
- Confirm reduced card effectiveness.
- Check failure sound effects.

Priority: Low

Estimation: 1 hrs

User story #26 Audio settings

As a user

i want to adjust audio settings

so that I can customize my experience in the game.

Validation:

- Verify volume controls work.
- Confirm settings are saved.
- Check real-time audio updates.

Priority: Medium

Estimation: 4 hrs

User story #27 Credits Screen

As a user

i want to view the game credits

so that I can see the development team and references.

Validation: • Verify credits screen access. • Confirm developer names and references appear. • Check return button functionality.	Priority: Medium Estimation: 3 hrs
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User story #28 Progression System As a user i want to improve as I progress through the game so that I feel rewarded for continuing to play.	
Validation: • Verify leveling system. • Check unlockable content. • See sadistics of wins and losses.	Priority: High Estimation: 9 hrs

User story #29 Difficulty Selection As a user i want to choose the game difficulty so that I can play according to my skill level.	
Validation: • Verify skill cards remain after defeat. • Confirm normal cards reset correctly. • Check restart system functionality.	Priority: High Estimation: 8 hrs

User story #30 Partial Progress Retention As a user	
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i want to keep part of my progress after losing
so that I do not have to restart completely from zero.

Validation:

- Verify difficulty options are available.
- Confirm enemy and math difficulty changes.
- Check selected difficulty is applied correctly.

Priority: Medium

Estimation: 8 hrs

User story #31 Pause Menu

As a user

i want to pause the game during gameplay
so that I can temporarily stop playing when needed.

Validation:

- Verify the game pauses correctly.
- Confirm pause menu appears.
- Check that the pause is not available while in combat.

Priority: Medium

Estimation: 6 hrs